

Toon Shader Pack

Requirements

- Tested with Unity 2022.3 LTS and Unity 6.0 LTS.
- Built-in or URP. HDRP is not supported.
- URP usage requires the Universal Render Pipeline package and an active Universal Render Pipeline Asset. The package does not automatically convert a Built-in project to URP.
- Mesh normals are required for lighting, rim light, and outlines. The Vertex Alpha shading source also requires vertex colors.

Quick Start

Paths are relative to `Assets/HaniJahanDesign/Shaders/Toon/`.

1. Open the matching demo scene:
 - Built-in: `Scenes/Shaders_Toon_Demo_BuiltIn.unity`
 - URP: `Scenes/Shaders_Toon_Demo_URP.unity`
2. Assign a preset from `Materials/BuiltIn/` or `Materials/URP/` to a mesh.
3. Adjust it in the custom material inspector.

To make a material from scratch, select `Hani Jahan Design/Toon/Built-in` or `Hani Jahan Design/Toon/URP` from its **Shader** menu.

Surface

Base Map provides the surface texture. **Tint** multiplies the texture, toon color, and alpha. The shader renders as an opaque material, so alpha can affect the calculated color but does not make the surface transparent.

Toon Shading

Enable **Toon Shading** and choose a **Shading Source** to produce a value from dark to light:

Source	Description
Main Light	Uses the main light direction and the Scene Lighting controls.
World Up	Maps downward-facing to upward-facing world-space normals.
Object Up	Maps downward-facing to upward-facing object-space normals.

Source	Description
World Height	Creates a gradient from the vertex's world-space height.
Object Height	Creates a gradient from the vertex's object-space height.
View Angle	Uses a Fresnel-style value based on the viewing angle.
Vertex Alpha	Reads the mesh vertex color's alpha channel.
Mask Texture	Reads the red channel of Mapping Mask , including its tiling and offset.

World Height and Object Height evaluate $\text{height} * \text{Height Scale} + 0.5$. Under **Advanced**, **Shading Offset** shifts the selected source and **Shading Contrast** biases it after the offset.

Next, choose how **Color Mode** converts that value into color:

- **Color Ramp:** samples a horizontal ramp texture, with dark on the left and light on the right.
- **Smooth Two Color:** blends continuously from **Shadow Color** to **Lit Color**.
- **Hard Two Color:** separates the two colors at **Shadow Size**. Set **Shadow Softness** to zero for a crisp boundary or raise it slightly to reduce aliasing.
- **Color Steps:** makes 2–8 evenly spaced bands between the two colors. **Step Softness** controls the boundaries between bands.

Disable Toon Shading to use the Base Map and Tint without authored toon colors. Rim and Scene Lighting can still affect the result.

Scene Lighting

When enabled, Scene Lighting reads the pipeline's main light direction and ambient spherical harmonics. It is intentionally lightweight and does **not** sample realtime or baked cast-shadow maps.

- **Ambient Strength:** controls how much ambient lighting brightens the authored surface color.
- **Wrap:** moves directional light around the sides of the model.
- **Shadow Falloff:** reshapes the main-light response.
- **Light Strength:** scales the main-light response.
- **Direction Override:** under **Advanced**, supplies a custom XYZ light direction. Leave XYZ at zero to follow Unity's main light.

Wrap, Shadow Falloff, Light Strength, and Direction Override apply when **Main Light** is the shading source. Ambient Strength can affect every shading source.

Rim Light

Rim light adds color near silhouettes based on the angle between the surface normal and camera:

- **Rim Color:** color and alpha multiplier for the effect.
- **Rim Power:** controls the rim's width; higher values make it narrower.
- **Rim Intensity:** controls brightness.

The final output is clamped, so strong rim settings can reach solid white or the maximum available color channel.

Outline

Outline renders an expanded, back-facing copy of the mesh before the surface:

- **Outline Color:** solid outline color.
- **Outline Width:** expansion along each vertex normal, measured in object-space units. Scale and mesh topology therefore affect the apparent width.
- **Outline Depth Offset:** shifts the expanded hull in clip-space depth to help with overlap artifacts. Start at zero and use small adjustments only when needed.

Hard edges, split normals, open meshes, very small details, and non-uniform scaling can create gaps or uneven outline thickness. Outline is a geometry-based inverted hull, not a screen-space outline, and does not add outlines between colors on the same surface.

Troubleshooting

- **Pink material:** use the shader matching the active pipeline. For URP, install/configure URP and assign a Universal Render Pipeline Asset; then check shader compile errors in the Console.
- **Flat or unexpected Main Light shading:** enable Scene Lighting, confirm the scene has an enabled main directional light, and reset Direction Override XYZ to zero. This shader does not receive cast shadows.
- **Ramp appears reversed or uniform:** place dark colors on the ramp's left and light colors on its right; verify the texture import and wrap settings.
- **Mask Texture has no effect:** choose Mask Texture as the Shading Source and put grayscale data in the texture's red channel.
- **Vertex Alpha has no effect:** confirm that the mesh contains varying vertex alpha values; a normal texture's alpha is not used.
- **Height gradient is misplaced:** choose world height for a scene-wide gradient or object height for a gradient that follows the object, then adjust Height Scale and Shading Offset.
- **Outline is missing:** enable Outline, use a mesh with valid normals, increase Outline Width for the object's scale, and check that Outline Color contrasts with the surface.

- **Outline is uneven:** apply or normalize object scale where practical and inspect the mesh normals. An inverted hull follows vertex normals and object-space scaling.

Known Limitations

- The shader is opaque and does not provide transparent surface blending.
- Scene Lighting uses main-light direction and ambient lighting but does not receive realtime or baked cast shadows, additional lights, reflection probes, or physically based highlights.
- The outline is an object-space inverted hull. Its apparent width varies with object scale, distance, projection, mesh normals, and topology.
- The shaders require shader model 3.0-capable targets.

Support

Publisher: Hani Jahan Design

Use the support contact on this asset's Unity Asset Store publisher page, or join the [HJD Discord](#) for feedback and community help.